Advanced NLP & Ethical AI

Sentiment Analysis & Explainable AI (XAI)

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B3B Advanced AI Course – Cohort 3
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SESSION FLOW

Today's Roadmap

01 Recap: From Generation to Understanding

02 Why Sentiment Analysis Is Harder Than It Looks

03 Lexicons & Bag-of-Words: First Attempts

04 From Statistics to Neural Sentiment

05 Why Models Must Explain Themselves

06 Intrinsic Explanations

07 Post-Hoc Explanations I: Attention & Saliency

08 Post-Hoc Explanations II: LIME & SHAP

09 Ethical AI: Bias, Fairness & Accountability

10 Wrap-up & Takeaways

Recap: From Generation to Understanding

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From “Can It Generate?” to “Can We Trust It?”

- Question 1: Sentiment Analysis
  - can a model understand what a person feels from their words?
  - This is classification, not generation — map text to a label (positive / negative / neutral), not produce new text.
- Question 2: Explainable AI (XAI)
  - A model decides on a sentiment, a loan approval, a medical flag
  - Can we understand WHY, and can we trust that reasoning is fair?
- This is the ethical heart of deploying any AI system in the real world, not just a technical afterthought.

Three Different Jobs for Three Different Models

Generation
produce new,
plausible content

→

Understanding
classify or extract
meaning from text

→

Explanation
justify a decision
for a human to audit

Same underlying toolbox with tokenizers, embeddings, attention, Transformers
But aimed at three different jobs. Today we build the second and third.

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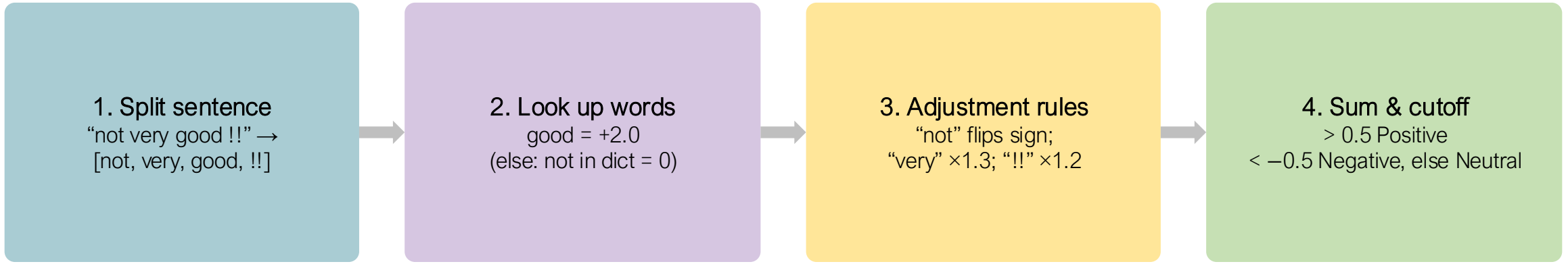
Just Count the Positive and Negative Words… Right?

- The obvious first approach:
  - Keep a dictionary of “good” words (great, love, amazing)
  - Keep a dictionary of “bad” words (terrible, hate, awful).
  - Count which side wins.
- This is roughly how the very first sentiment tools worked
  - it's a genuinely useful starting point, not a strawman.
  - We'll build the real algorithm, by hand, next.

The Lexicon Algorithm, Step by Step

- A “lexicon” is just a dictionary: a big list pairing each word with a hand-picked score for how positive or negative it sounds.

Not very good !

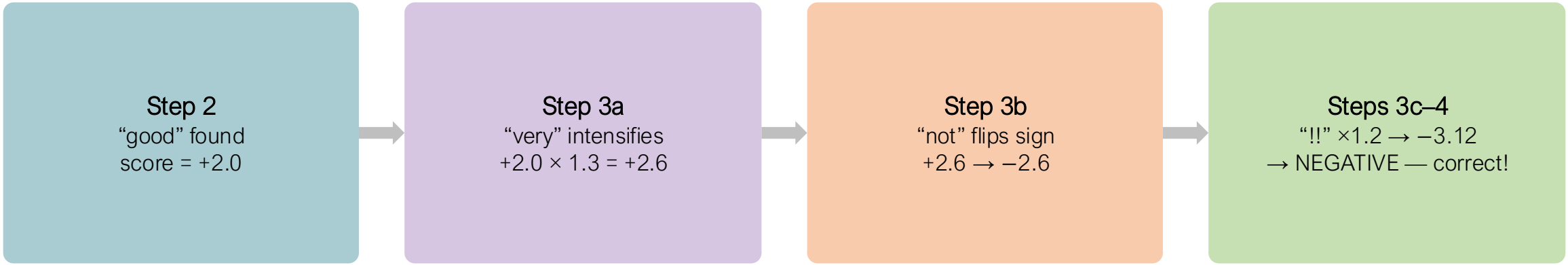


-1.0 x 2.0 x 1.3 x 1.2 = -3.12

-3.12 < – 0.5 -> negative

Running the Algorithm, by Hand: Yet another example

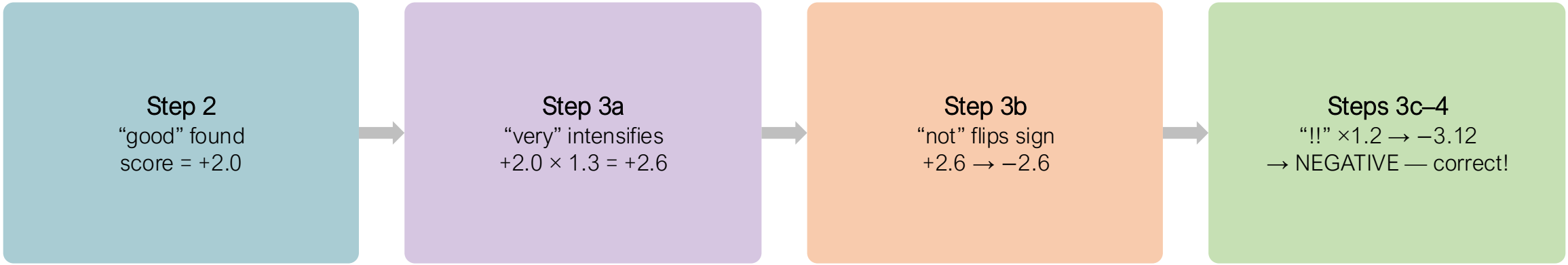
- “The movie was not very good!!”



This simplified recipe is genuinely close to how VADER (Hutto & Gilbert, 2014) works
A real, widely-used lexicon tool
Dictionary + a handful of hand-written rules for negation, intensifiers, and punctuation.

Running the Algorithm, by Hand: Yet another example

- “The movie was not very good!!”



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Does it always work?

It Does not always work: Same Algorithm, Wrong Answer

- “Oh great, another delay.”

Step 2: “great” found → score = +3.0. No negation word nearby, no “not,” nothing to flip it.
Step 4: +3.0 > 0.5 → classified POSITIVE — but the speaker is clearly annoyed. WRONG.

Why it's “not sufficient”

every rule in Step 3 is a LOCAL, hand-written patch
it has no global knowledge that *delays are annoying*
so it can never detect *irony* like this, and there are infinite ways to be sarcastic.

Try It Yourself: The World's Most Dramatic Soup Review

"This soup was so hot it violated three laws of physics, filed a restraining order against my tongue, and is now legally banned in four banned in four states. 10/10, would cry again."

- Run it through our algorithm: "banned" and "violated" score negative, "10/10" scores positive — the dictionary has no idea this is a JOKE about how good the soup was. A human laughs; the lexicon algorithm just sees contradictory word scores and shrugs.

Moral of the story: if a sentence needs a sense of humor to parse, word-counting was never going to survive it.

Try It Yourself: The World's Most Dramatic Soup Review

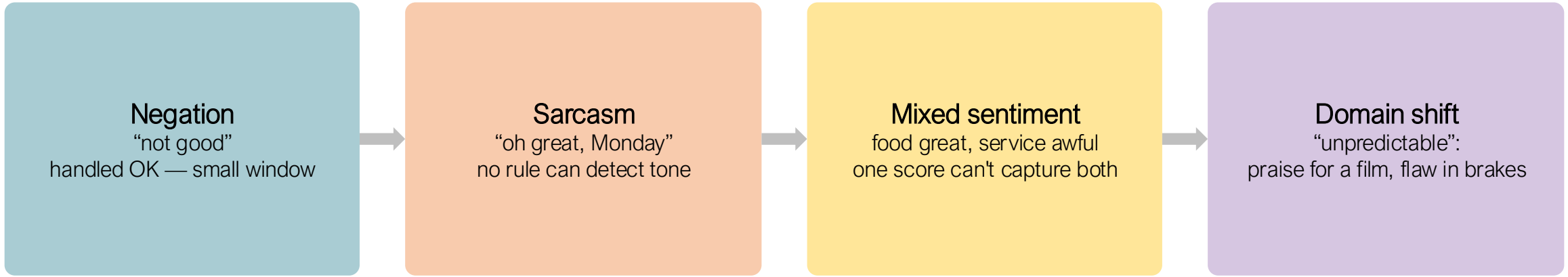
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Moral of the story: if a sentence needs a sense of humor to parse, word-counting was never going to survive it.

Everything is going perfectly. 🙃

Four Ways Word-Level Rules Get Fooled



Before We Move On…

Q In the lexicon algorithm, why couldn't ANY amount of extra hand-written rules fully detect sarcasm?

Sarcasm depends on knowing that delays are generally unwelcome — real-world knowledge the dictionary and its local rules simply rules simply don't contain. Sarcasm has effectively unlimited forms, and hand-writing a rule for each is a losing battle. This
This motivates letting a model LEARN patterns from data instead — next up.

Lexicons & Bag-of-Words: First Attempts

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What Is a “Bag of Words,” Really?

- Picture this: take every word in a sentence, rip it out of order, and dump all the words into a bag — all you have left is a pile of words, and you can count how many of each you have.

Sentence	great	movie	boring	acting
“ great movie”	1	1	0	0
“ boring acting”	0	0	1	1

Each sentence becomes a row of numbers (a “vector”)
notice “movie” and “boring” never touch;
the model has no idea they were ever in the same sentence, or in what order.

Bag-of-Words + Machine Learning, Step by Step



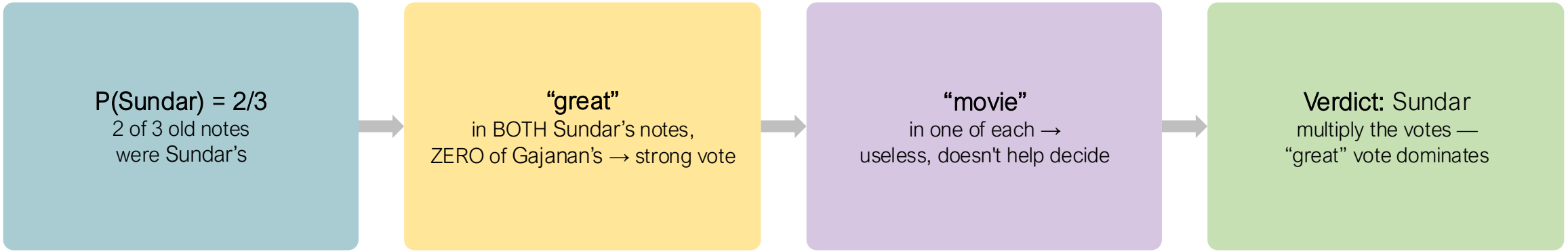
Naive Bayes, Explained Like You're a Detective

- You're a detective. A note arrives: “Interesting acting.”
  - Was it left by a HAPPY reviewer or a GRUMPY one?
  - You've read this detective’s previous notes before, so you can make an educated guess.
  - Say previously for 90% cases “interesting!” means not very good.
  - That is Baye’s rule
- Bayes' Rule FLIPS a question you can't answer directly into one you can:
  - instead of “given these words, which reviewer?” (hard),
  - Ask “given each reviewer, how likely are these exact words?” (easy — just check their old notes).
- “Naive” simply means the algorithm treats every word as if it contributes independently

$$P(\text{happy} \mid \text{words}) \propto P(\text{happy}) \times P(\text{word}_1|\text{happy}) \times P(\text{word}_2|\text{happy}) \times \ldots$$

Naive Bayes, By Hand

- Two potential authors: Sunny Sundar (always upbeat) and Grumpy Gajanan (never satisfied).
- Old files: “great movie” → Sundar, “great acting” → Sundar, “boring movie” → Gajanan. New note: “great movie” — Sundar or Gajanan?



Nobody told our detective that “great” is Sundar’s writing
it figured that out purely by noticing the pattern in old case files.
That's the entire trick behind Naive Bayes.

Bag-of-Words Loses Word Order: Can be problematic

- Two very different opinions — identical bag of words:

"Food not bad, service not good."
mildly positive about food,
negative about service



"Food not good, service not bad."
negative about food,
mildly positive about service

Both sentences contain the EXACT same words, same counts.
Bag-of-Words literally cannot tell them apart, because it throws away order

Naive Bayes vs. VADER: Which One is better?

- Surprising fact: our “learn from data” Naive Bayes is often MORE primitive than the “hand-written rules” VADER, in one specific way.

Naive Bayes	VADER (lexicon)
Zero built-in language knowledge — no idea what “ not” or “ very” even mean	Humans hand-coded negation, intensifiers, punctuation & emoji rules
Only handles negation/intensifiers IF it saw similar examples in training	Handles “ not very good” or “ :)” correctly on day one, zero training data
Needs labelled training data to learn anything at all	Needs no labelled data — works immediately on a new domain
Strength: improves with more data, adapts to new domains	Weakness: frozen rules — can't adapt without a human rewriting them

- On a small dataset, VADER often wins.
- If there are millions of labelled reviews, Naive Bayes usually wins instead
- Naive Bayes starts picking up patterns no human thought to hand-code.

Lexicons & Bag-of-Words: What to Remember

- ✓ Lexicon methods (like VADER) are dictionary + hand-written rules — fast and need no training data, but frozen and blind to anything beyond their rules (like sarcasm).

- ✓ Bag-of-Words throws away word order on purpose, turning a sentence into a simple word-count vector; Naive Bayes Naive Bayes learns which words predict which label directly from data.

- ✓ Both families share the same blind spot: neither one reads words IN ORDER — exactly the wall we hit next.

Lexicons vs. Learning Trivia

A brand-new startup has only 50 labelled customer reviews but needs sentiment analysis working THIS WEEK. Lexicon (VADER) or Naive Bayes — which do your use, and why?

VADER — it needs no training data at all and already handles negation, intensifiers, and punctuation out of the box. 50 examples is nowhere near enough to train a reliable Naive Bayes classifier.

MODULE 4 OF 10

From Statistics to Neural Sentiment

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The Common Thread — and What's Still Missing

- Look back: lexicons use hand-rules, Naive Bayes learns from data
    - BOTH treat a sentence as an unordered pile of words.
- So “not bad, actually good” and “not good, actually bad” looked identical to Bag-of-Words
    - neither approach ever reads a sentence in order.
- The fix: give the model a way to read words IN SEQUENCE and build up context as it goes
    - Use embeddings and attention

Recap: What's a Word Embedding

- A word embedding is just a list of numbers assigned to a word
  - e.g. “great” might become [0.8, –0.2, 0.5, …]
  - It's a coordinate, a point in space.
- These numbers are learned so that words with SIMILAR meaning end up as NEARBY points.
  - “Great” and “wonderful” land close together; “great” and “concrete” land far apart.

✨ INSIGHT Why this matters for sentiment: if a model learns “wonderful” is positive, and “fantastic” sits right next to it in this number-space, the model already has a head start on “fantastic” too, even without seeing it in training.

Recap: What's Self-Attention

Situation: You read “The service was slow, but I'd still call it good”

- Your eyes jump back to “slow” and “good” when you hit “still”
- As if each word “listens” to the others that matter most for understanding it.

- Self-attention computes exactly that
  - for every word against every other word it is a weight capturing “how much should I pay attention to this word, when interpreting that word?”

This is precisely the missing ingredient: unlike Bag-of-Words, a self-attention model can connect “not” all the way to “good” even with several words in between — order and context are no longer thrown away.

History: from Embeddings to Fine-Tuned Transformers

- 2014–2017: feed word embeddings into an RNN/LSTM that reads the sentence in order, one word at a time, carrying forward a running memory — finally sensitive to word order.
- 2018+: fine-tune a pretrained BERT (built from stacked self-attention layers) for classification — add one small classification head on top of a summary token's output, then train on labelled reviews.
- This is Supervised Fine-Tuning again — take a model that already understands language broadly, and nudge it toward one specific job: producing a sentiment label instead of a generated sentence.

$$\text{predicted label} = \text{highest-scoring option out of } \{\text{positive, negative, neutral}\}$$

Aspect-Based Sentiment Analysis (ABSA): One Sentence, Two Opinions

- Remember the "mixed sentiment" trap — "Food was great, service was awful"
- It broke plain word-counting because ONE score can't capture TWO opinions. Aspect-Based Sentiment Analysis (ABSA) predicts sentiment for each of the topic mentioned.

Food — Positive
"The food was great,"
evidence: "great"

Service — Negative
"... service was painfully slow."
evidence: "painfully slow"

Mixed sentiment is what ABSA is for: a restaurant owner needs to know their SERVICE is the problem, not just that reviews average out to "mixed."

How Do Today's LLMs Do Sentiment Analysis?

- Today's largest LLMs (GPT-4-class, Claude-class) often skip fine-tuning ENTIRELY for sentiment analysis.
  - Just ask “what is the sentiment”: this is called zero-shot or few-shot classification.
- “Few-shot” means showing 2–3 example (sentence → label) pairs directly in the prompt before the real question
  - Called in-context learning, an active research area (which examples to choose?)

$$\text{predicted label} = \text{argmax over } v \in \{\text{positive, negative, neutral}\} \text{ of } \Pr(v \mid \text{prompt})$$

That's the exact same next-token prediction objective of LLMs, just with the output restricted to a tiny set of label words.
Trade-off: prompting is flexible, but a fine-tuned classifier is usually far cheaper at scale.

Two Ways to Get Sentiment Today

```
# (A) Classic fine-tuned classifier
from transformers import pipeline
classifier = pipeline("sentiment-analysis")
classifier("The food was great, but service was painfully slow.")
# [{'label': 'NEGATIVE', 'score': 0.81}]

# (B) Zero-shot via a large language model
prompt = """Classify as positive, negative, or neutral.
Review: "The food was great, but service was painfully slow."
Label:"""
llm_response = llm.generate(prompt)
# "negative"  -- no fine-tuning, just next-token prediction
```

Same underlying idea both times — predict a label from text — but (B) reuses the raw generative engine from Lecture 1 with zero task-specific training.

Which Sentiment Approach, When?

Need	Approach
Zero labelled data, running TODAY, must explain every point	Lexicon (VADER)
Modest labelled data (hundreds–thousands), lightweight & fast	Bag-of-Words + Naive Bayes
Large labelled dataset, best accuracy on a fixed task at scale	Fine-tuned Transformer (BERT)
Per-topic opinions, not one overall label, from a mixed review	Aspect-Based Sentiment Analysis
Little/no labelled data, need it THIS AFTERNOON	Zero-/few-shot LLM prompting

Real systems often chain these: an LLM prompt to bootstrap labels fast, then a fine-tuned model to run cheaply at scale.

Sentiment Analysis: What to Remember

- ✓ Bag-of-Words' blind spot — no word order — is exactly what embeddings + attention fix, by letting the model read in model read in sequence and connect distant words.

- ✓ Fine-tuning a pretrained Transformer's summary token is today's classic standard; ABSA solves the “mixed sentiment” trap by scoring each topic separately.

- ✓ Cutting-edge LLMs often skip fine-tuning entirely — zero-/few-shot prompting reuses the same next-token prediction objective from Lecture 1.

Classifier Trivia

Q Why can a self-attention model succeed where Bag-of-Words fails on “not bad, actually good” vs. “not good, vs. “not good, actually bad”?

Self-attention reads the sentence in order and computes, for every word, how strongly it should connect to every other word — so “not” can attach specifically to the word right next to it. Bag-of-Words throws away position entirely, so both sentences produce the exact same word-count vector.

Why Models Must Explain Themselves

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What if Your Sentiment Classifier Makes a Mistake…

- Picture the fine-tuned sentiment detection transformer, deployed at a company inbox. A furious customer writes a scathing complaint — and the model tags it “Neutral.” A human never sees it flagged as urgent.
- For a lexicon or Naive Bayes, we could answer “why” directly — trace the word scores. For a fine-tuned Transformer with 100+ million parameters, we genuinely can't just “look.”

This gap between a model's power and our ability to inspect it is the subject of the rest of this lecture

Same Question for Many High-Stakes Places

- The same “why did it decide that?” question shows up everywhere a model's decision affects a real person:



Sentiment

A furious complaint gets tagged “neutral” and silently ignored



Lending

A loan is denied. “40M numbers decided together” isn't a real answer



Medicine

A scan is flagged low-risk. Which features drove that call?



Hiring

A resume is rejected. Was it skills — or something it shouldn't use?

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This is the foundational motivation of eXplainable AI or XAI

XAI Not Just For Understanding: It's Often the Law

EU GDPR, Article 22, 2018

- The EU's GDPR (2018), Article 22, grants individuals a right to meaningful information about automated decisions that significantly affect them — an applicant CAN legally demand an explanation.

This regulation turns explainability from “nice to have” into a hard requirement for any organization deploying models on real people — sentiment systems included, wherever they gate a real decision.

Accuracy vs. Interpretability — Two Different Axes



Accuracy

How often the model's prediction matches the true answer — a single measured number.



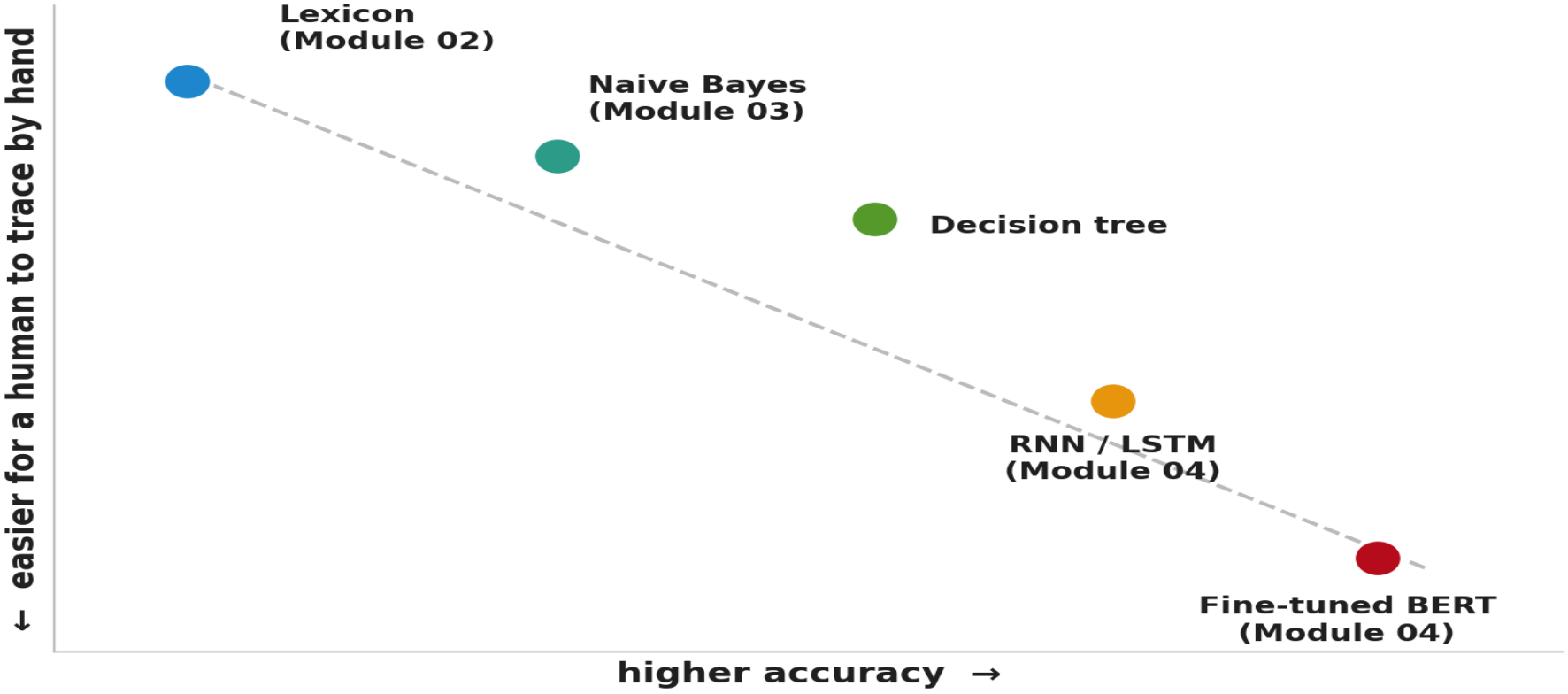
Interpretability

How easily a HUMAN can trace, step by step, why the model produced a specific output.

These are DIFFERENT axes, not two names for the same thing — a model can be highly accurate yet totally opaque, or transparent yet mediocre. Nothing forces them to move together.

The Accuracy ↔ Interpretability Trade-off

- In PRACTICE (not by mathematical law), the models we've built this lecture tend to line up along a diagonal:



The rest of this lecture is about clawing back interpretability WITHOUT giving up the accuracy of the right side
that's the entire point of XAI.

Before We Move On…

Q Why can't we just always use the most interpretable model (like a lexicon) and skip this whole problem?

Interpretable models often can't capture the messy, context-dependent patterns that make sentiment hard in the first place (recall sarcasm, Module 02). For high-stakes decisions where accuracy genuinely matters, we need a way to keep the powerful model AND still get an explanation — that's exactly what Modules 06–08 build.

Intrinsic Explanations

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Why Intrinsic Explanations Work

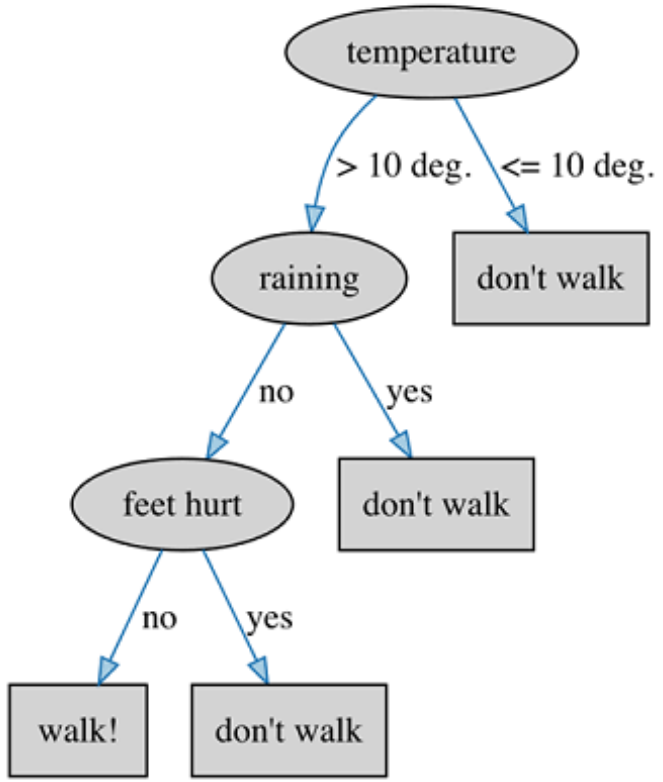
- “Intrinsic” means the model is built simply enough that its computation and its explanation are literally the SAME object — nothing extra to add afterward.
- A decision tree is nothing but a sequence of yes/no questions.
    - To make a prediction, you walk down the tree
    - That exact walk, written down, IS the explanation.

Contrast a fine-tuned Transformer:

- Its “computation” is 100+ million numbers multiplying together
- Not something a human can walk through step by step.
- That's the whole reason post-hoc methods (Modules 07–08) had to be invented.

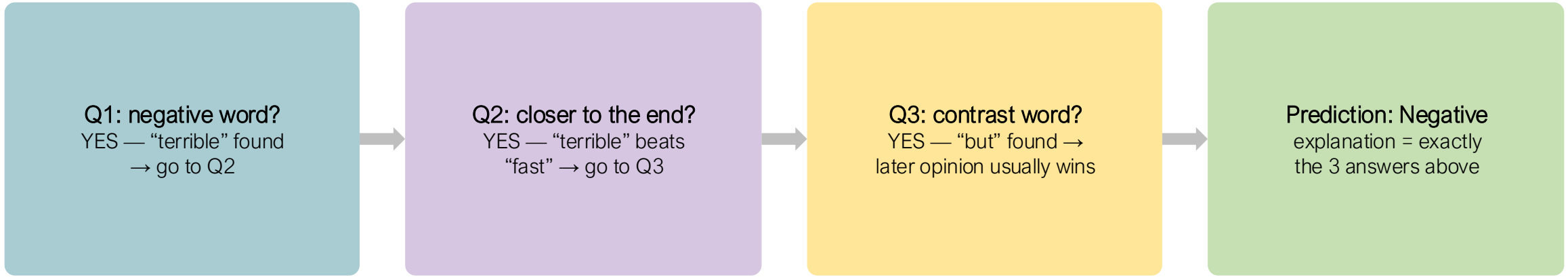
Intrinsic Explanation: Example

- Since
  - Temperature > 10 deg
  - AND raining = yes
  - Don’t walk



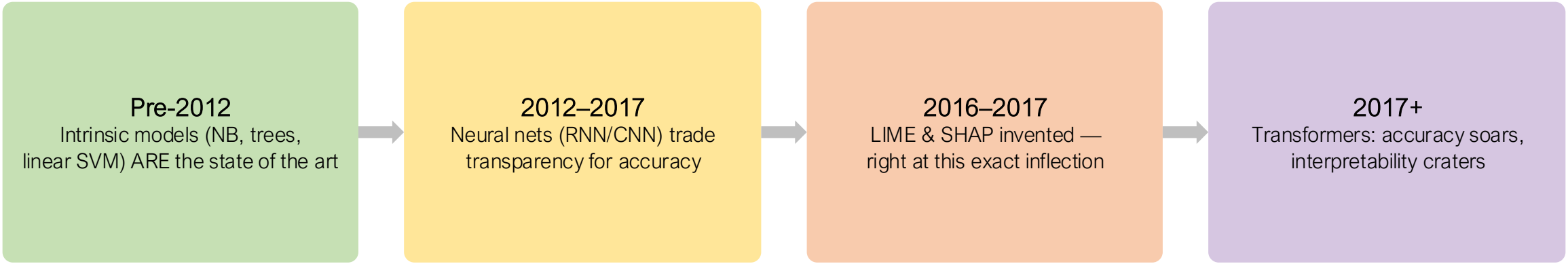
A Tiny Decision Tree for Sentiment

- “This phone is fast but the battery life is terrible.” — walk it through a 3-question tree:



The walk through the tree IS the complete explanation.

When Did Intrinsic Stop Being “Enough”?



Notice this historical timing isn't a coincidence:
post-hoc XAI methods emerged exactly when models got too complex for intrinsic explanations to keep up.

Intrinsic Explanations: What to Remember

- ✓ Intrinsic explanations work because the model's decision procedure IS simple enough for a human to walk through by hand — no approximation needed.

- ✓ Before 2012, intrinsic models were both the most accurate AND the most explainable option available — there was no trade-off to make.

- ✓ Post-hoc XAI methods (LIME 2016, SHAP 2017) were invented specifically because Transformer-era accuracy gains made intrinsic explanations impossible to keep up with.

Intrinsic Trivia

Q Why can't we just build a decision tree big enough to be as accurate as a fine-tuned Transformer, and get and get intrinsic explanations for free?

At that size the PATH through the tree becomes just as long and tangled as a neural network's computation — no longer something a human can meaningfully trace by hand. Intrinsic interpretability requires staying SMALL and SIMPLE, which is precisely the accuracy trade-off from Module 05.

MODULE 7 OF 10

Post-Hoc Explanations I: Attention & Saliency

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Back to the Black Box

- decision tree was intrinsically explainable — but that transparency came at a real accuracy cost.
- For the fine-tuned Transformer— the one that's actually accurate enough to deploy
    - we need a DIFFERENT strategy
    - keep the black box exactly as it is, and build a separate tool that PROBES it after the fact.
    - That's “post-hoc” explanation.

What Was “Attention”, Again?

Analogy: picture a party where everyone is technically listening to everyone, but the SECOND someone says “so anyway, I heard…” every head swivels toward them. Self-attention computes exactly that, mechanically.

- Tempting idea for explanation: after the model predicts, look at which words got the HIGHEST attention weights, and treat those as “this is what the model focused on.”

It's an appealing way to generate explanation
the weights already exist inside the model as a side-effect of the computation, no extra work needed.

Attention on a Real Sentence

- “The food was great, but the service was painfully slow.” — shaded by how much attention the model paid to each word



- “painfully slow” and “great” light up — which matches intuition. But does high attention always mean “causally important”?

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The honest caveat: Jain & Wallace (2019), “Attention Is Not Explanation,” showed you can often find a completely different attention pattern that produces the SAME prediction — high attention is a hint, not proof.

What's a “Gradient”?

Analogy: standing on a foggy hillside, you can't see the whole landscape, but your feet can feel which direction the ground slopes upward. The “gradient” is exactly that felt slope: which direction makes the function increase fastest, and how steeply.

- For explanation, we ask: “if I nudge this ONE word's numbers slightly, how much does the model's confidence in its prediction change?” A big change = a steep slope = that word mattered a lot.

Saliency: A Direct Sensitivity Measurement

- Saliency computes the gradient of the predicted class's score with respect to each input word's embedding
  - Words with a large gradient are ones where a small nudge would most change the model's confidence.
- Unlike attention, this is a direct measurement of sensitivity, not a side-effect of the computation
  - Much closer to a real causal signal for “this word mattered.”

$$\text{saliency}(\text{word}_i) = \left| \partial(\text{model output}) \,/\, \partial(\text{embedding of word}_i) \right|$$

Attention & Saliency: What to Remember

- ✓ Attention weights are a free-but-imperfect signal — they show where the model “looked,” which isn't always what caused the decision.
- ✓ Saliency (gradients) directly measures how sensitive the output is to each word — a more direct, if costlier, causal causal probe.
- ✓ Both are post-hoc: they explain an already-trained black box after the fact, without changing how it works.

Attention vs. Saliency Trivia

Q Why is gradient-based saliency generally considered more trustworthy than raw attention weights?

Saliency directly measures how sensitive the output is to a small change in each word's representation — a genuine causal probe. Attention weights only tell you where the model “looked,” which research (Jain & Wallace, 2019) has shown doesn't always correspond to what actually drove the final decision.

Post-Hoc Explanations II: LIME & SHAP

01 Recap: From Generation to Understanding

02 Why Sentiment Analysis Is Harder Than It Looks

03 Lexicons & Bag-of-Words: First Attempts

04 From Statistics to Neural Sentiment

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06 Intrinsic Explanations

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08 Post-Hoc Explanations II: LIME & SHAP

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10 Wrap-up & Takeaways

LIME: Interrogate the ML Black Box With “What If”

Analogy: picture the world's most persistent toddler interrogating a chef. “What if no tomato?” “What if no salt?” “What if no salt AND no carrots?” After enough “what if” questions, they've worked out which ingredient actually matters. That's LIME, minus the baby tantrums.

- LIME (Ribeiro, Singh & Guestrin, 2016): to explain one prediction, generate many perturbed versions of that input (remove or mask a few words at a time), and see how the model's output changes each time.
- Fit a simple, interpretable model (e.g. linear regression) to those (perturbation → output) pairs — LOCALLY, just around this one input.

The weights of that local linear model become the feature-importance explanation for this one prediction.

LIME, Written Out as Pseudocode

- 1. Pick the input x — e.g. x = “great acting, boring plot”; let f(x) = the black box's prediction about sentiment
- 2. Generate N perturbed samples — z_i = x with a random subset of words removed/masked
- 3. Get the black box's prediction for each — $y_i = f(z_i)$, just run the model
- 4. Weight each sample by closeness to x — w_i = similarity(z_i, x)
- 5. Fit a simple weighted linear model — $g(z) = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \ldots$
- 6. Read off the explanation — each β IS the importance of that word for this prediction

LIME, By Hand — One Word at a Time

- Sentence: “great acting, boring plot” → model says 62% positive. Mask one word at a time and re-run:

Perturbation	Input to model	Score	LIME's inferred effect
original	great acting, boring plot	0.62	—
mask “great”	[MASK] acting, boring plot	0.35	−0.27 → “great” pushes positive
mask “boring”	great acting, [MASK] plot	0.81	+0.19 → “boring” pushes negative
mask “acting”	great [MASK], boring plot	0.60	−0.02 → “acting” barely matters

This is LIME: perturb, observe, and read off which words moved the needle
no need to open up the model's internals at all.

SHAP: Explanation Rooted in Game Theory

Analogy: three friends make one pizza and split the bill fairly by VALUE added. Imagine every possible order they could've joined the kitchen, and kitchen, and average each person's improvement across all orders — that average is their fair share. That's SHAP.

- SHAP (Lundberg & Lee, 2017) applies Shapley values — a 1953 cooperative-game-theory solution — treating each WORD as a “player” and the model's output as the “payoff.”
- For each word, SHAP averages its marginal contribution across every possible order the words could be “added” to the sentence.

Unlike LIME, SHAP values are guaranteed to add up exactly to (prediction − average prediction)
a mathematical fairness guarantee LIME doesn't make.

SHAP, Written Out as Pseudocode

- 1. Pick x and the word to credit — e.g. explain “boring” in “great acting, boring plot”
- 2. Consider every ORDER words could join — for each ordering π, build up the sentence one word at a time
- 3. Measure “boring”'s marginal effect — contribution = f(so far + “boring”) − f(so far without it)
- 4. Average over ALL orderings — SHAP value = average of all contributions
- 5. Repeat for every word — each word gets its own fairly-averaged SHAP value
- 6. Check the guarantee — sum of all SHAP values = f(x) − f(average input), exact by construction

LIME and SHAP in Practice

```
import shap
from transformers import pipeline

classifier = pipeline("sentiment-analysis", return_all_scores=True)
explainer = shap.Explainer(classifier)

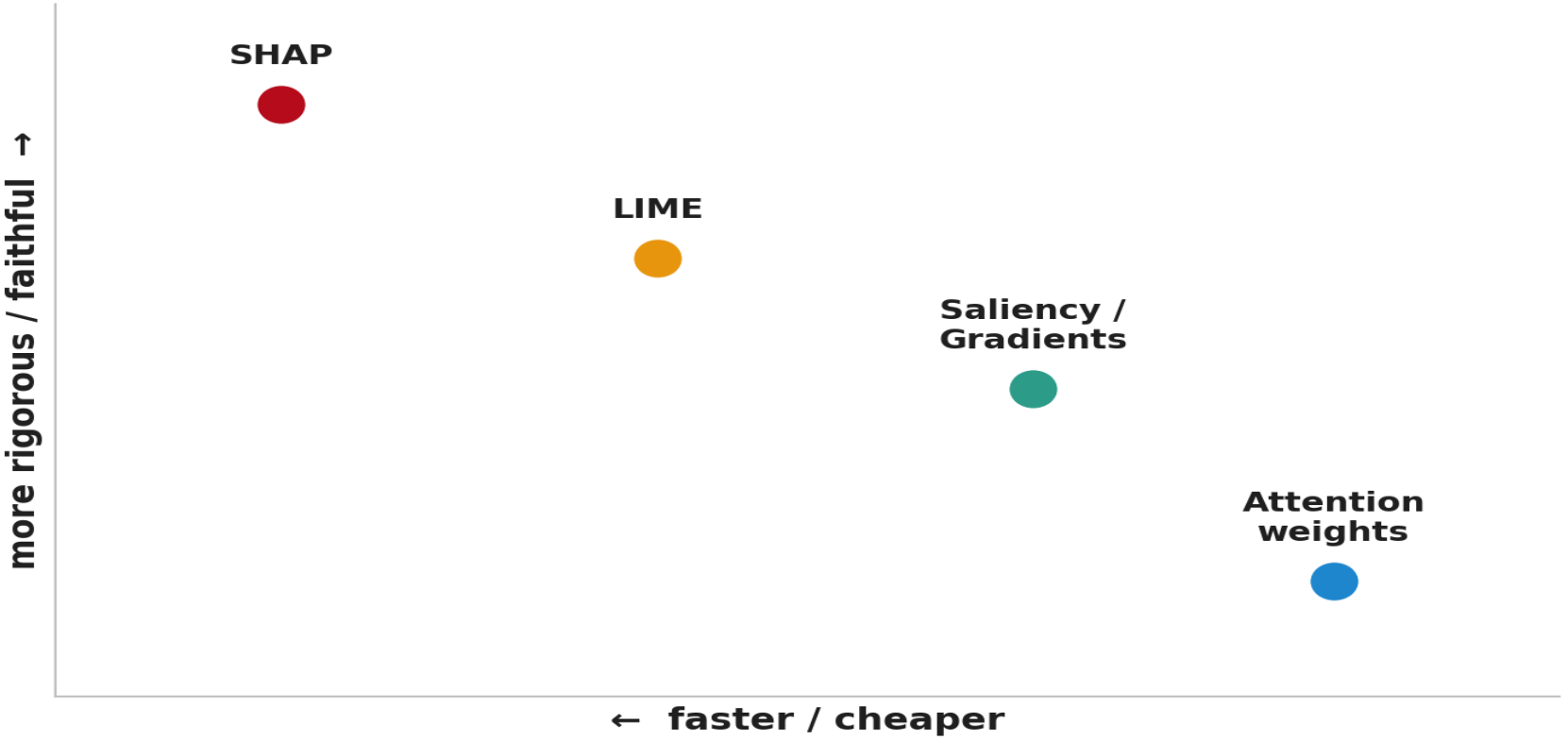
shap_values = explainer(["great acting, boring plot"])
shap.plots.text(shap_values[0])

# Output (conceptually):
#   great   ██████  +0.29  (pushes POSITIVE)
#   boring  ██████  -0.24  (pushes NEGATIVE)
```

Same idea as our by-hand LIME table — SHAP just computes it more rigorously, and ships as a one-line drop-in for any Hugging Face pipeline.

The Landscape of Post-Hoc Methods

- Plotting every technique from Modules 07–08 by speed vs. faithfulness (how rigorously it reflects the model's TRUE reasoning):



✨ INSIGHT Attention is free but weakest; SHAP is the most principled but the most expensive to compute — LIME and saliency sit in between.

Which Technique, When?

Need	Question	Answer
SOME signal, instantly, for free	Model already has attention?	Attention — but sanity-check it
Fast, cheap, any differentiable model	Have access to gradients?	Saliency / gradient-based methods
Explain ONE prediction, model-agnostic	Can afford a few hundred re-runs?	LIME
Rigorous, fair, guaranteed credit	Can afford the heaviest compute?	SHAP
Full audit trail, some accuracy loss OK	High-stakes / regulated decision?	Intrinsic model (Module 06)

LIME & SHAP: What to Remember

- ✓ LIME perturbs an input, observes the effect, and fits a simple local model — fast, intuitive, but only an approximation. approximation.

- ✓ SHAP treats each feature as a player in a cooperative game and computes a fair, mathematically guaranteed split of credit — slower, but more principled.

- ✓ Both are post-hoc and model-agnostic — they work on any classifier, from Naive Bayes to fine-tuned BERT, without without needing access to its internals.

LIME vs. SHAP Trivia

Q In our by-hand LIME table, why did masking “acting” barely change the prediction?

The score only moved from 0.62 to 0.60 — “acting” isn't carrying much sentiment on its own; it's a neutral, descriptive noun. Contrast “great” and “boring,” whose removal swung the score by 0.27 and 0.19 — exactly the kind of signal LIME is designed to surface.

MODULE 9 OF 10

Ethical AI: Bias, Fairness & Accountability

01 Recap: From Generation to Understanding

02 Why Sentiment Analysis Is Harder Than It Looks

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We Can Now See Inside the Model. So What?

- So far, we have decision trees, attention, saliency, LIME, SHAP
- We can pry open a black box and see WHY it decided something.
- But seeing why is only step one. What happens when we open the box and don't like what we find — a hiring model relying on gender, a sentiment model scoring certain names more harshly?

what do we DO with an explanation once we have it — is where explainability meets ethics.
Now we will talk about exactly that handoff.

What Is “AI Ethics,” Actually?

- AI ethics is the study of how to design and deploy AI systems that treat people fairly, respect their privacy and autonomy, and are ACCOUNTABLE for their real-world impact — not just accurate on a test set.



Fairness
resume scoring differently by gender or race



Privacy
trained on private messages, can leak them back out



Autonomy
a feed engineered to be addictive, overriding intent



Accountability
no one can explain why a decision was made

In this module we will specifically focus on FAIRNESS

When “Accurate on Average” Isn't Good Enough

ProPublica, COMPAS, 2016

Amazon hiring model, 2018

Kiritchenko & Mohammad, *SEM 2018

- ProPublica's 2016 investigation of COMPAS, a US recidivism-risk tool, found it flagged Black defendants as “high risk” at nearly twice the twice the false-positive rate of white defendants — despite similar overall accuracy.
- Amazon scrapped an internal hiring model in 2018 after discovering it had taught itself to penalize resumes containing the word “women's” — learned from a decade of historically male-dominated hiring data.
- Kiritchenko & Mohammad (2018) found several commercial sentiment systems scored otherwise-identical sentences more negatively when they contained African-American-associated names versus European-associated names.

Connecting the Stories to Their Root Cause

Historical bias
training data reflects real, historical inequities (Amazon)

→

Sampling bias
some groups are under-represented in the data

→

Label bias
annotators' assumptions enter the “ground truth” (COMPAS)

None of these require a malicious engineer
bias is usually the default outcome of training on real-world data, unless someone actively checks for it.

How XAI Helps Catch Bias and become an ethical tool

- Run SHAP or LIME on a hiring model's rejections: if a name, a zip code, or a gendered word carries large positive/negative weight, that's a visible, auditable red flag — not a hidden pattern buried in millions of weights.

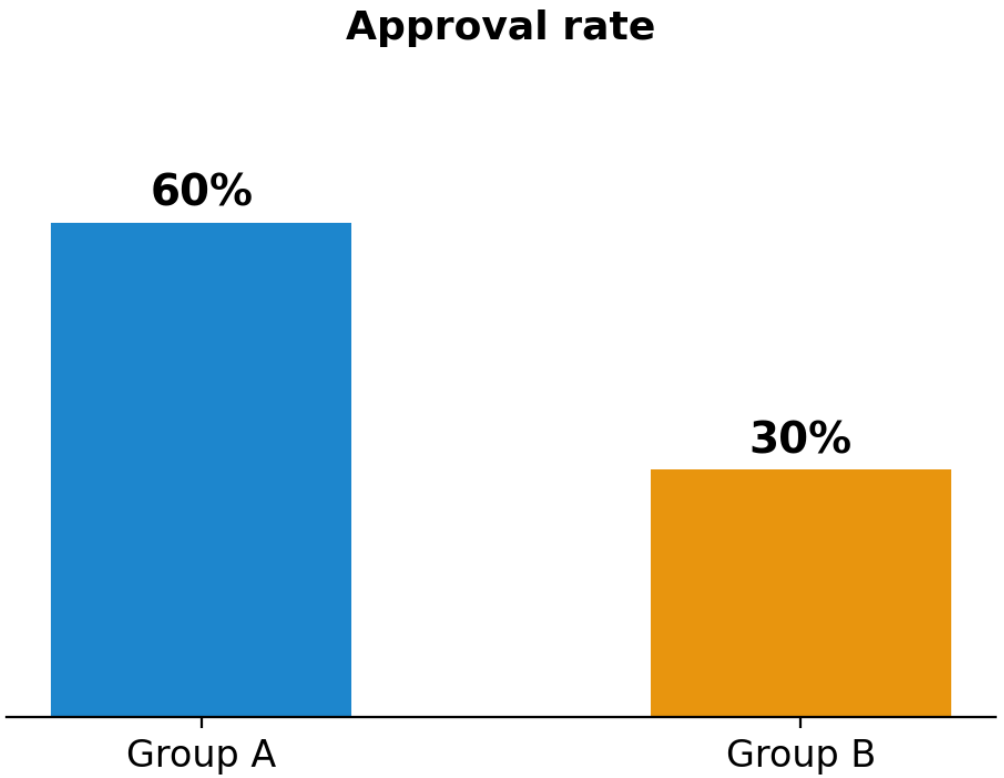
This is exactly the COMPAS and Amazon scenarios:
- Identify the right explanation tool pointed at the right cases
- The biased feature is visible before deployment, not after a public investigation.

Fairness Definition 1: Demographic Parity

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b) \quad \text{for every pair of groups } a, b$$

- $\hat{Y}$ is the model's decision (1 = approved/flagged positive); A is a protected attribute. Demographic parity says: the RATE of positive decisions must be equal across groups.

60% ≠ 30% → demographic parity VIOLATED, regardless of whether either group actually deserved approval.

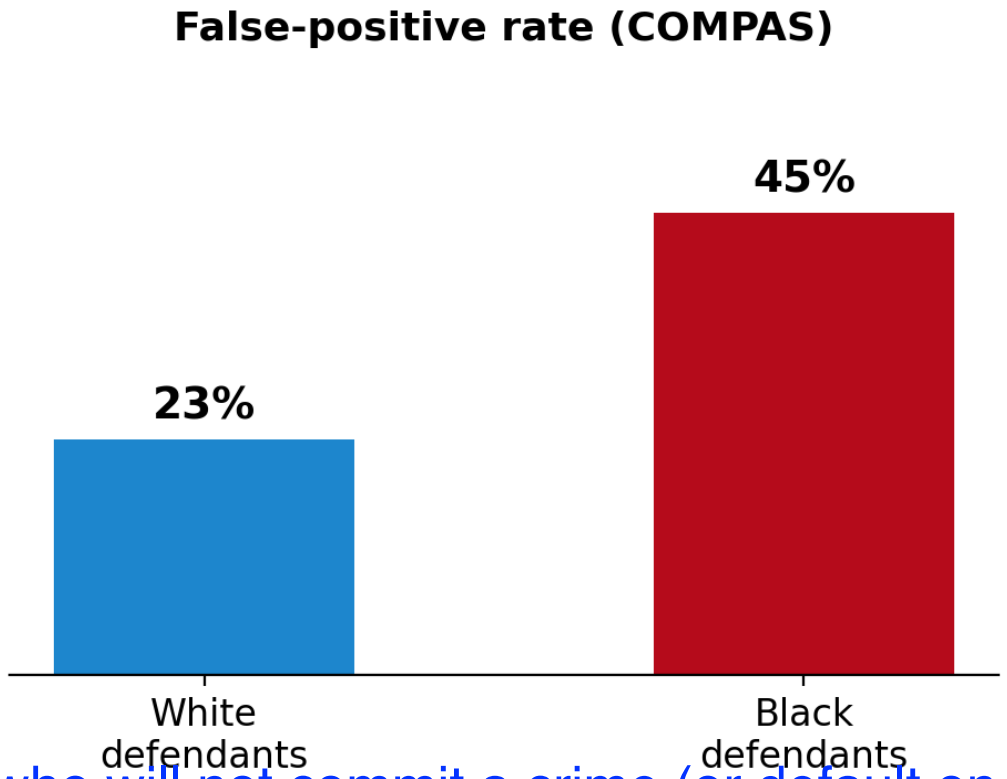


Fairness Definition 2: Equalized Odds

$$P(\hat{Y}=1 \mid Y=y, A=a) = P(\hat{Y}=1 \mid Y=y, A=b) \quad \text{for both } y = 0 \text{ and } y = 1$$

- Y is the TRUE outcome. Equalized odds says:
  - TAKE PEOPLE WHO ARE ACTUALLY THE SAME (same true Y)
  - the model's error rate must be equal across groups
  - This is stricter than demographic parity.

23% ≠ 45% → equalized odds VIOLATED — this is literally the ProPublica finding. A model can satisfy one while violating the other — other — there is no single “fair.”



Equal False Positive Rates (y=0) : Out of all people who will not commit a crime (or default on a loan)

Equal True Positive Rates (y=1): Out of all people who will commit a crime (or successfully pay back a loan)

Documentation as a Governance Tool

- Model Cards (Mitchell et al., 2019): a short, standardized document shipped alongside a model — intended use, evaluated subgroups, known limitations.
- Datasheets for Datasets (Gebru et al., 2018): the same idea for training data — collection source, demographics, known gaps.
- EU AI Act (2024): sorts systems into risk tiers — minimal (spam filters), limited (chatbots, disclosure), high (hiring, credit — mandatory documentation & bias testing), unacceptable (social scoring — banned).

None of these replace XAI
they're the paperwork that makes an organization actually DO the explaining, consistently, every time, not just when a journalist asks.

Where Does Each Ethics Tool Actually Work?

Stage	Tool
BEFORE training: is the data itself skewed?	Datasheets for Datasets
AFTER training, ONE prediction: used a bad feature?	LIME / SHAP on individual rejections
AFTER training, WHOLE model: outcomes skewed?	Demographic Parity check
AFTER training, WHOLE model: mistakes skewed?	Equalized Odds check
BEFORE deployment: prove due diligence?	Model Cards + EU AI Act risk tier

XAI tools catch bias in ONE decision; fairness metrics catch it across the WHOLE model; documentation catches it across the WHOLE organization, over time.

Ethical AI: What to Remember

- ✓ AI ethics is broader than bias — fairness, privacy, autonomy, and accountability are four distinct concerns, each with real-world failure cases.
- ✓ Bias is usually the default outcome of training on real-world data (historical, sampling, or label bias), not malicious intent.
- ✓ Demographic parity and equalized odds are both valid, both mathematically precise, and can conflict — there's no single definition of “fair.”

Ethical AI Trivia

Q A loan model achieves demographic parity (equal approval rates across two groups). Does that guarantee it guarantee it also satisfies equalized odds?

No. Demographic parity only checks that overall approval RATES match — it says nothing about whether those approvals are equally accurate for people who would actually repay versus default. The two definitions are mathematically independent.

MODULE 10 OF 10

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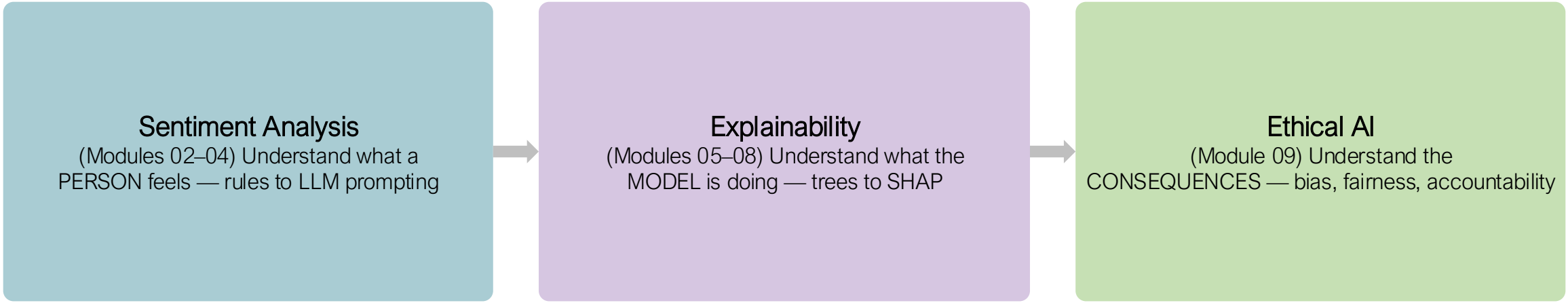
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Understanding People, Models, and Consequences



Key Takeaways

Topic	One-line takeaway
Sentiment Analysis	Every method — rules, Naive Bayes, Transformers, LLM prompting — solves the same problem with progressively less hand-engineering and more learning from data.
Why Explanation Matters	Accuracy and interpretability are different axes that trade off in practice; intrinsic models ARE their own explanation, post-hoc tools approximate one for black boxes.
LIME, SHAP & Friends	Perturb-and-observe or fairly split credit via game theory — pick the technique using speed/rigor/stakes, not habit.
Ethical AI	Bias is usually the default outcome of real-world data; demographic parity and equalized odds are both valid, both precise, and can conflict.

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